

CHEN XU

Hangzhou, China • chenxu1098@gmail.com • [GitHub](#) • [Portfolio](#) • [Codeforces \(International Master\)](#)

EDUCATION

New York University, Tandon School of Engineering

Expected May 2030

B.S. in Computer Science • Class of 2030

- Competitive Programming: Data Structures, Dynamic Programming, Tree & Graph Algorithms, Probability, Combinatorics.
- Languages: English and Mandarin (both native)

HONORS & COMPETITIONS

- Gold Medal:** CCPC&ICPC Shandong Regional, 2026
- Gold Medal:** ICPC Xi'an Regional Contest, 2025
- Silver Medal:** ICPC Asia East Continent Final, 2024
- Silver Medal:** ICPC Asia Chongqing Regional, 2025
- Codeforces International Master. USACO Platinum Division

EXPERIENCE

Computer Vision Engineer

Feb – Apr 2026

- Sole developer of a CCD machine-vision system that measures battery-electrode segment widths in active production on **30+ machines across 5 battery manufacturers**, carried through 50+ iterative releases driven by real plant feedback
- Owned the full product: chessboard calibration, electrode measurement and QC, real-time TCP integration with the inspection line's control system, plus a PySide6 GUI and a JSON-output CLI for upstream automation
- Built the edge-detection pipeline from primitives: banded $|Sobel-x|$ gradient response $\rightarrow$ non-max-suppression multi-peak finding $\rightarrow$ per-peak RANSAC line fit (parameterized $x = m \cdot y + b$ so near-vertical edges stay stable), with brightness-step and angle filters to reject foil wrinkles and scratches as false edges
- Designed a binary-search algorithm that finds the minimal relaxation of three coupled detection gates (peak threshold, RANSAC confidence, angle tolerance) to recover the expected edge count while admitting the fewest false edges; calibrated against ground truth at $R^2 > 0.995$

PROJECTS

Usage-Aware GPU Memory Oversubscription — HAMi (open-source Kubernetes GPU virtualization)

2026

- Diagnosed that HAMi oversubscribes VRAM against *requested* quota while ignoring its own per-container real-usage signal, a blind over-commit that triggers CUDA OOM when memory-tight jobs co-locate
- Closed the loop in the Go scheduler so admission becomes $real_used + request \leq physical\ capacity$, safely reclaiming idle GPU memory; gated behind an env flag with automatic fallback to stock behavior when the usage signal is unavailable
- Validated trace-driven simulation and live on a real Kubernetes/V100 cluster: at **zero crashes**, raised utilization **55% $\rightarrow$ 78% (1.4 $\times$)** vs. an optimally-tuned baseline and **2 $\times$ (28% $\rightarrow$ 56%, 2 $\times$ the admitted pods)** vs. the fixed-scaling production default

Agent Toolchain & LLM Infrastructure

2026

- Built agent tooling on MCP. Authored custom MCP servers exposing tools to LLM agents, and drove multi-step coding workflows through agent harnesses (Claude Code) including tool-use loops and subagent orchestration
- Hands-on with the LLM serving and agent stack: API integration, KV-cache behavior, prompt and context engineering, and computer-use agents (UI-TARS Desktop)

BookaCook — iOS Application (\$20K Hackathon Grand Prize, 100+ downloads)

2025

- Shipped full-stack iOS recipe platform; built Python web crawlers to scrape, dedupe, and index **1,200+ cookbooks and 1M+ recipes** into SQLite backend
- Integrated multi-model AI pipeline: TikTok API ingestion $\rightarrow$ Whisper for speech-to-text $\rightarrow$ GPT-4o for ingredient recognition from images $\rightarrow$ Llama 3 8B for structured recipe generation

TECHNICAL SKILLS

Languages: C++, Python, Go, Swift, SQL

ML & AI: PyTorch, MCP, agent harnesses, LLM API integration, KV-cache, prompt & context engineering

Systems & Infra: Kubernetes, GPU virtualization, CUDA/GPU memory, Linux/CLI, Git

Computer Vision: OpenCV, camera calibration, RANSAC line fitting

Mathematics: Probability, statistics, linear algebra, algorithms & data structures